

TITLE PAGE

- **Problem Statement ID-** SIH26034
- **Problem Statement Title-** Software System to check compliance of Packaged Commodities under Legal Metrology (Packaged Commodities) Rules, 2011 by scanning products, images and labels.
- **Theme-** ?Theme as listed on the SIH portal?
- **PS Category-** Software
- **Team ID-** ?Team ID?
- **Team Name (Registered on portal)-** ?Team name as registered?

NET QUANTITY

500 g

M.R.P. Rs 145.00

Mfd: 03/2026

Packer: name & address

I letter height

MEASURED

millimetre estimate with an uncertainty interval

ABSTAINED

reason recorded; no violation inferred

AI and computer vision assist observation. The officer records the legal determination.

Offline-first field assistant: capture the package, extract the declarations, check them against the rules of that date, measure letter height only when the evidence supports it ? and abstain, on the record, when it does not.

NiyamDrishti ? offline-first Legal Metrology screening assistant for packaged commodities: capture ? extract ? check ? measure-or-abstain ? evidence-linked report

? Proposed Solution (Describe your Idea/Solution/Prototype)

| Detailed explanation

- ? Android app for a Legal Metrology inspection: the officer captures the package faces, the app extracts the declarations, evaluates a versioned rule pack and assembles the record ? entirely on-device.
- ? Character-height declarations add a calibrated path: a printed co-planar reference frame turns a phone photo into a millimetre estimate.
- ? The output is a candidate finding with its evidence, not a verdict.

| How it addresses the problem

- ? Every finding links to its source crop, raw and normalised observation, rule version and engine version, so a reviewer can re-trace it.
- ? Unreadable or unsuitable evidence returns PHYSICAL_SIZE_NOT_ESTABLISHED instead of a confident violation.
- ? Contradictions across faces ? two different MRPs, for instance ? are preserved and flagged, never silently resolved.
- ? No connectivity is needed in a shop or godown; sync is optional.

| Innovation and uniqueness of the solution

- ? Evidence-bounded measurement: every millimetre figure carries an uncertainty interval and a guard-band decision.
- ? Fail-closed gates ? calibration, geometry, image quality, segmentation, burst spread, uncertainty ? run before any number is shown.
- ? Face and glyph coverage is pre-registered and hash-checked, so nothing is selected after a result is seen.

Field officer workflow

CAPTURE

package faces

IDENTIFY

product & faces

EXTRACT

declarations (OCR)

CHECK

rule pack, by date

MEASURE

or abstain

EVIDENCE

crop + hash link

REVIEW

officer confirms

REPORT

canonical JSON + PDF

?Don?t guess.?

An unreadable or unsuitable condition must never become a confident violation. The system states what it could not establish.

Where the boundary sits

AI / CV assists

detection, OCR, boundary estimation, gating

System states

MEASURED with an interval · or ABSTAINED with a reason code

Officer decides

reviews the evidence and records the legal determination

Technologies to be used

Client	Android (Kotlin), offline-first; encrypted local store; optional idempotent sync
Vision	on-device OCR with face and declaration-line detection; every crop retained as evidence
Measurement	four-marker co-planar fiducial, homography rectification, sub-pixel 50 % linearised-luminance ink boundary, model-fitted extremes
Rules	signed, versioned rule pack selected by inspection date; deterministic three/four-valued logic
Uncertainty	component budget ? combined u_c ; $k = 1.645$, reported as nominal and not yet calibrated
Evidence	canonical JSON plus an offline PDF, hash-linked; role-based review
Reference build	pure-Python measurement engine with no third-party numerical stack; 352 automated tests; deterministic re-runs

Built and tested today: the measurement engine, its benchmark and the validation tooling. The Android client is the implementation target of this proposal, not a finished product.

Methodology ? how a photo becomes a bounded millimetre estimate

Known physical reference



Controlled capture



Geometry correction



Printed boundary estimation



Physical size estimate



Uncertainty

Fail-closed gates choose the branch: calibration, geometry, image quality, segmentation, burst spread, uncertainty. A gate that cannot be satisfied abstains.

The measurand is documented explicitly ? visible printed-ink extent perpendicular to the fitted baseline ? and labelled a screening proxy, not a statutory definition.

MEASURE

ABSTAIN



OFFICER REVIEW

Validation status ? what is evidence today, and what is not

Controlled synthetic benchmark

- ? 76 runs, 10 of 10 pre-registered criteria PASS
- ? max arithmetic error 0.008 mm on ideal renders
- ? median burst repeatability SD 0.0025 mm

Abstention behaviour, same benchmark

- ? 11 of 11 targeted unsafe cases abstained
- ? 0 false-clear of 4 undersized; 0 false-accuse of 4 compliant
- ? measure rate 1.00 under safe conditions

Physical laboratory validation

- ? P1?P7 acceptance criteria pre-registered and frozen
- ? reference procedure, glyph map and panel selection committed
- ? laboratory validation pending ? no physical claim is made

? Feasibility of the idea**Technical**

- ? The measurement pipeline is implemented and reproducible: re-runs are bit-identical and 352 automated tests pass.
- ? No third-party numerical stack, so the same arithmetic ports to the device.
- ? Measured cost about 1.3 s per 12 MP frame single-threaded ? a burst takes seconds.

Operational

- ? The field kit is a phone plus a printed co-planar reference frame; no bench instrument travels to the shop.
- ? Offline by design: the inspection completes with no network.
- ? Training is a short protocol card, because the app refuses conditions it cannot handle.

Legal and ethical

- ? Screening assistance: the officer records the determination, the app records the evidence.
- ? Rule-version aware and date-selected, so a past inspection stays evaluable under the rules of its own date.
- ? Local-only storage by default; sync is opt-in.

Pre-committed stop conditions

- | | |
|-----------------------|---|
| C1 fails | the pipeline arithmetic is wrong ? fix before any physical work |
| P1 fails | no usable ground truth ? no accuracy claim of any kind |
| P2 or P6 fails | the measurement claim is downgraded |
| P5 fails | the capture protocol changes, never the threshold |

Written down before the experiment, so a disappointing result cannot be answered by moving a threshold.

? Challenges, risks and how each is handled**Curved, flexible or embossed packs**

Declared out of scope. Out-of-plane print is invisible to a single view ? measured 0.185 mm error with every gate green ? so flatness is a fixture requirement and the app abstains instead.

No trustworthy ground truth

An independent reference procedure is pre-registered: a 2400 dpi scan as primary and a measuring-microscope cross-check, with criterion P1 gating every accuracy claim.

The interval is not yet calibrated

$k = 1.645$ is reported as nominal. P7 collects empirical coverage as calibration evidence and can never be shown as a validated interval.

OCR or transcription error

Every field keeps its source crop for officer review, and cross-face contradictions are surfaced rather than resolved.

Reference equipment access

The minimum setup is documented ? 2400 dpi scanner, certified length standard, measuring microscope ? with machine-checked readiness gates before capture.

Which glyphs the rule counts

The threshold is configurable per rule version, and the measurand is documented as a proxy pending confirmation with the administering authority.

? Potential impact on the target audience**Primary user: the Legal Metrology field officer or inspector.**

- ? A structured pass over the package faces, with coverage tracked instead of remembered.
- ? Evidence captured once at the point of inspection, so declarations are not re-typed later.
- ? Conditions that cannot be read are recorded as such: the officer is told what was not established.
- ? A reproducible record ? same inputs, same output ? with rule and engine versions attached.
- ? Supervisory review becomes re-tracing evidence instead of reconstructing a visit.
- ? Works in a shop, godown or checkpoint with no connectivity.

Deliberate limits, stated up front**Scope**

planar printed faces, controlled capture

Role

screening assistance, not adjudication

Decision

the officer determines compliance

Stating this on the record is part of the design: a screening tool that overstates its reach is worse than one that refuses a case.

What ?evidence-linked? means here

Each link is stored, so a supervisor can re-trace a finding to the pixel it came from and to the rule text that was in force.

? Benefits of the solution**Consumer protection (social)**

- ? Declarations are screened consistently rather than from recall of the rule text.
- ? Weak evidence produces an abstention, so a compliant packer is less likely to be wrongly accused.

Administrative

- ? Uniform, machine-readable records with a lineage that runs from the finding back to the pixel.
- ? Coverage and abstention rates become visible quantities a supervisor can act on.

Economic (qualitative)

- ? Commodity hardware and an offline workflow, so no per-site infrastructure is needed.
- ? Open, versioned rule packs keep maintenance inside the department.

Environmental

- ? A paperless inspection record produced on the device, printed only when a case requires it.

Measured as the pilot runs, not asserted here: face coverage per inspection, abstention rate by reason code, and the share of findings with a complete evidence chain.

No time-saving, accuracy or cost figure is claimed: none has been measured in the field yet.

Regulatory and metrological references

- 1. Legal Metrology (Packaged Commodities) Rules, 2011**
Mandatory declarations on pre-packaged commodities, and their size and legibility. Department of Consumer Affairs.
consumeraffairs.nic.in
- 2. Legal Metrology Act, 2009**
Enabling statute for the Rules and for enforcement powers.
consumeraffairs.nic.in
- 3. Consolidated Rules text used to build the rule model**
The 2011 Rules with amendments, as hosted for public reference.
bombayhighcourt.gov.in/bhc/libweb/legislation/rulec/LegalMetrologyPackagedCommoditiesRules,2011.pdf
- 4. OIML R 79 ? Labelling requirements for prepackaged products**
International reference for declaration and legibility requirements.
oiml.org/en/publications
- 5. JCGM 100:2008 (GUM) ? Evaluation of measurement data**
The method used for the component uncertainty budget and coverage factor.
bipm.org
- 6. Problem statement SIH26034**
Ministry of Consumer Affairs, Food & Public Distribution ? Department of Consumer Affairs.
sih.gov.in

The open question we do not paper over

Open question	which glyphs and faces the rule counts, and whether ink extent is the right reading of letter height
Our position	documented as a screening proxy with a configurable threshold per rule version, to be confirmed with the administering authority before any compliance wording is finalised

Our own engineering evidence (this project)

- Frozen protocol and criteria**
P0_PROTOCOL.md and P0_CRITERIA.md ? the capture protocol and the pre-registered P1?P7 acceptance bands.
- Independent reference method**
P0_REFERENCE_PROCEDURE.md ? scanner-primary, microscope cross-check, and the independence rules between them.
- Pre-registered coverage**
glyph_map.json, 400 glyph instances; glyph_selection.json, 20 panels ? both derived by rule and hash-checked.
- Readiness audits**
Reference setup, flatness control and equipment inventory, each with a machine-checked gate.
- Reference implementation**
Pure-Python measurement engine, a 76-run synthetic benchmark with its analysis output, and 352 automated tests.

Repository

github.com/dinamsingh/Setu · branch **niyamdrishti-phase0**

Holds the measurement engine, the synthetic benchmark and its analysis output, the frozen protocol and criteria, the pre-registered coverage files, the readiness audits and the test suite. Private repository ? access on request.

Every synthetic figure quoted in this deck is reproducible from the repository. Physical laboratory validation has not been performed and is not claimed anywhere in this deck.